



QUESTIONS NUMERICAL METHODS FOR PDES

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Question 1. *What type of PDE is the Poisson equation? Why? What is a well-posed problem? Why is the solution to the Poisson problem unique?*

Solution 1. Let $\Omega \subseteq \mathbb{R}^d$ be an open and bounded domain and $\mathbf{A} : \Omega \rightarrow \mathbb{R}^{d \times d}$, $\mathbf{b} : \Omega \rightarrow \mathbb{R}^d$ and $c : \Omega \rightarrow \mathbb{R}$ be smooth enough functions. Consider the differential operator

$$(1) \quad \mathcal{L}u(\mathbf{x}) = -\mathbf{A} : \text{Hess}[u](\mathbf{x}) + \mathbf{b}(\mathbf{x}) \cdot \nabla u(\mathbf{x}) + c(\mathbf{x})u(\mathbf{x}).$$

Under the assumption that $\mathbf{A} \in C^2$ we can, by Schwartz' theorem, assume that $\mathbf{A}(\mathbf{x})$ is symmetric for all $\mathbf{x} \in \Omega$. We say that

- (a) $\mathbf{A}$ is *elliptic* in $\mathbf{x}$ if $\mathbf{A}(\mathbf{x})$ is positive definite, i.e., all eigenvalues of $\mathbf{A}(\mathbf{x})$ are positive.
- (b) is *hyperbolic* at $\mathbf{x} \in \mathbb{R}^d$ if $\mathbf{A}(\mathbf{x})$ has one negative and $n - 1$ positive eigenvalues.
- (c) is *parabolic* at $\mathbf{x} \in \mathbb{R}^d$, if $\mathbf{A}(\mathbf{x})$ positive semi-definite but not definite (i.e., one eigenvalue is zero and $\text{rank}(\mathbf{A}|\mathbf{b}) = n$)

We then see that we have $\Delta u = \mathcal{L}u$ by taking $\mathbf{A} = \mathbf{I}$. Since $\mathbf{I}$ is positive definite, Δ is elliptic.

Definition. A differential equation with initial conditions is *well posed* if the solution *exists* and is *unique* and *continuously depends on the initial datum*.

Otherwise we call the problem *ill posed*.

Lemma 1.1. *Suppose that $u \in C^2(\Omega) \cap C^0(\overline{\Omega})$ solves*

$$(2) \quad \begin{cases} -\Delta u = f & \text{in } \Omega \\ u = g & \text{on } \partial\Omega \end{cases}$$

with f and g continuous. Then u is unique.

Proof. Suppose that u_1, u_2 are both solutions. Then $v = u_1 - u_2$ is harmonic thus by the maximum principle

$$|v(\mathbf{x})| \leq \max_{\mathbf{y} \in \partial\Omega} |v(\mathbf{y})| = 0.$$

Lemma 1.2. *Suppose that $u_i \in C^2(\Omega) \cap C^0(\overline{\Omega})$ solves*

$$\begin{cases} -\Delta u_i = f & \text{in } \Omega \\ u_i = g_i & \text{on } \partial\Omega \end{cases}$$

for $i = 1, 2$ with f, g_i continuous. Then

$$|u_1(\mathbf{x}) - u_2(\mathbf{x})| \leq \max_{\mathbf{y} \in \partial\Omega} |g_1(\mathbf{y}) - g_2(\mathbf{y})|.$$

Proof. Since $u_1 - u_2$ is harmonic, the maximum of $|u_1 - u_2|$ is attained at the boundary. □

Question 2. *What type of PDE is the heat equation? Why? What is a well-posed problem? Why is the solution to the heat equation unique?*

Solution 2. The heat equations reads

$$\begin{cases} \partial_t u(t, \mathbf{x}) - \operatorname{div}_{\mathbf{x}} [c(\mathbf{x}) \nabla_{\mathbf{x}} u(t, \mathbf{x})] = f(t, \mathbf{x}) & \text{for } \mathbf{x} \in \Omega, t \in (0, T), \\ u(t, \mathbf{x}) = 0 & \text{for } (t, \mathbf{x}) \in (0, T) \times \partial\Omega, \\ u(0, \mathbf{x}) = u_0(\mathbf{x}) & \text{for } \mathbf{x} \in \Omega. \end{cases}$$

If we rewrite this equation in the form (1) we get, using

$$\operatorname{div}(\mathbf{A}(\mathbf{x}) \nabla u(\mathbf{x})) = [(\underbrace{\nabla^\top \cdot \mathbf{A}}_{=:\operatorname{div}(\mathbf{A})^\top}) \nabla u + \mathbf{A} : \operatorname{Hess}[u]](\mathbf{x})$$

so we can write the equation as

$$\underbrace{\begin{bmatrix} 0 & \mathbf{0} \\ \mathbf{0} & \mathbf{A} \end{bmatrix}}_{:=\tilde{\mathbf{A}}} : \begin{bmatrix} \partial_{t,t}^2 u & \partial_t \nabla_{\mathbf{x}}^\top u \\ \partial \nabla_{\mathbf{x}} u & \operatorname{Hess}_{\mathbf{x}}[u] \end{bmatrix} + \begin{bmatrix} 1 \\ \operatorname{div}(\mathbf{A}) \end{bmatrix} \cdot \begin{bmatrix} \partial_t u \\ \nabla_{\mathbf{x}} u \end{bmatrix} = f$$

Since $\mathbf{A}$ has full rank and is positive semidefinite (iff $c(\mathbf{x}) > 0$) we obtain that

$$\operatorname{rank} \begin{bmatrix} 0 & \mathbf{0} & 1 \\ \mathbf{0} & \mathbf{A} & \operatorname{div}(\mathbf{A}) \end{bmatrix} = n + 1,$$

i.e., the heat equation is a parabolic equation.

Lemma 2.1. *The solution to the heat equation with $c \equiv 1$ is unique.*

Proof. First we define the quantity

$$E[u](t) := \frac{1}{2} \int_{\Omega} |u(t, \mathbf{x})|^2 d\mathbf{x}.$$

Using Green's first identity, which reads,

$$\int_{\Omega} v \Delta u d\mathbf{x} = - \int_{\Omega} \nabla u \cdot \nabla v d\mathbf{x} + \int_{\partial\Omega} v \frac{\partial u}{\partial \mathbf{n}} dS$$

we find

$$\begin{aligned} \frac{d}{dt} E[u](t) &= \int_{\Omega} u(t, \mathbf{x}) \underbrace{\partial_t u(t, \mathbf{x})}_{=\Delta u} d\mathbf{x} \\ &= - \int_{\Omega} |\nabla u|^2 d\mathbf{x} + \underbrace{\int_{\partial\Omega} u \frac{\partial u}{\partial \mathbf{n}} d\mathbf{x}}_{=0} \leq 0 \end{aligned}$$

from which we deduce that

$$E[u](t) \leq E[u](0) = \int_{\Omega} u_0(\mathbf{x}) d\mathbf{x}.$$

Now, u_1, u_2 be two solution of the heat equation. Then $v = u_1 - u_2$ satisfies the heat equation with $f = 0$, zero boundary-conditions and zero-initial condition. From the above estimate one deduces $v \equiv 0$ i.e., $u_1 \equiv u_2$. $\square$

Question 3. *Define the weak derivative and the Hilbert space $H^k(\Omega)$. Does $u(x) = |x|$ lie in $L^2(\Omega)$, $H^1(\Omega)$, or $H^2(\Omega)$ for $\Omega = (-1, 1)$? Why or why not?*

Solution 3. Let $u \in L_{\text{loc}}^1(\Omega)$ we say that u has α -th partial weak derivative $v^\alpha \in L_{\text{loc}}^1(\Omega)$ for a multiindex α if

$$\int_{\Omega} u \partial^\alpha \varphi d\mathbf{x} = (-1)^{|\alpha|} \int_{\Omega} v^\alpha \varphi d\mathbf{x} \quad \text{for all } \varphi \in C_c^\infty(\Omega).$$

If v^α and $\tilde{v}^\alpha$ are both weak derivatives of u we have that

$$\int_{\Omega} (v^\alpha - \tilde{v}^\alpha) \varphi d\mathbf{x} = (-1)^{|\alpha|} \int_{\Omega} (u - u) \partial^\alpha \varphi d\mathbf{x} = 0 \quad \text{for all } \varphi \in C_c^\infty(\Omega)$$

which implies that v^α is a.e. unique. Using integration by parts one also verifies that classical and weak derivatives coincide so we denote the weak derivative by $\partial^\alpha u$.

We now define for $k \in \mathbb{N}$

$$H^k(\Omega) := \{u \in L^2(\Omega) \mid \text{for all } |\alpha| \leq k, \partial^\alpha u \text{ exists and is in } L^2(\Omega)\}$$

if $k \geq 1$ and $H^0(\Omega) := L^2(\Omega)$.

We now consider the example $u(x) = |x|$ for $x \in (-1, 1) =: \Omega$. Obviously $u \in L^2(\Omega)$. Also for any $\varphi \in C_c^\infty(\Omega)$ we have

$$\begin{aligned} \int_{\Omega} |x| \varphi'(x) dx &= \int_0^1 x \varphi'(x) dx - \int_{-1}^0 x \varphi'(x) dx \\ &= \underbrace{\left[x \varphi(x) \right]_{x=0}^1}_{=0} - \int_0^1 \varphi(x) dx - \underbrace{\left[x \varphi(x) \right]_{x=-1}^0}_{=0} + \int_{-1}^0 \varphi(x) dx \\ &= - \int_{\Omega} (\mathbb{1}_{\mathbb{R}_+}(x) - \mathbb{1}_{\mathbb{R}_-}(x)) \varphi(x) dx. \end{aligned}$$

Hence, $H(x) := \mathbb{1}_{\mathbb{R}_+}(x) - \mathbb{1}_{\mathbb{R}_-}(x)$ is the weak derivative of $|x|$. We now show that H does not have a weak derivative. Assume that h is the weak derivative of H . Then it must hold that

$$\begin{aligned} - \int_{\Omega} h(x) \varphi(x) dx &= \int_{\Omega} H(x) \varphi'(x) dx \\ &= - \int_{-1}^0 \varphi'(x) dx + \int_0^1 \varphi'(x) dx \\ &= -2\varphi'(0) \end{aligned}$$

Take $\varphi(x) := \exp\{-\frac{1}{x^2-1/4}\}$ and extend it by 0 outside Ω and set $\phi_n(x) := \varphi(nx)$. Then plugging this in and using dominated convergence,

$$0 = - \lim_{n \rightarrow +\infty} \int_{\Omega} h(x) \phi_n(x) dx = -2 \cdot \lim_{n \rightarrow +\infty} \phi_n'(0) = -2 \cdot \lim_{n \rightarrow +\infty} n \varphi'(0) = +\infty,$$

which is a contradiction. Or we can just argue that H^1 functions must be continuous in 1-D. $\square$

Question 4. Consider the boundary value problem

$$(P) \quad \begin{cases} -u'' = f, & \text{in } \Omega = (0, 1), \\ u(0) = 0, & u'(1) = 0 \end{cases}$$

where $f \in C^0(\bar{\Omega})$. Derive its variational formulation (V). For $u \in C^2(\bar{\Omega})$ show that (V) is equivalent to (P).

Solution 4. The variational formulation of (P) which reads

find $u \in V$ s.t.

$$(V) \quad \int_0^1 u' v' dx = \int_0^1 f v dx \text{ for all } v \in V := \{v \in H^1((0, 1)) \mid v(0) = 0\}$$

is obtained using integration by parts. Note that we can assign values at the boundary because $v \in H^1((0, 1))$ has a continuous representant. Assume that $u \in C^2(\bar{\Omega}) \cap V$ is a solution to (V). Then for any $v \in V \cap C^1([0, 1])$ we have (using integration by parts)

$$\int_0^1 (f + u'') v dx - v(1) u'(1) = 0.$$

So taking $v \in H_0^1((0, 1))$ we obtain by the fundamental theorem of variational calculus (because $u'' + f$ is continuous) that $f \equiv -u''$. Lastly, taking $v(x) = x$ we obtain that

$$\begin{aligned} \int_0^1 u'(x) dx &= - \int_0^1 u''(x) \cdot x dx + \left[u'(x) \cdot x \right]_{x=0}^1 \\ &= \int_0^1 f(x) v(x) dx + \underbrace{\left[u'(x) \cdot x \right]_{x=0}^1}_{=u'(1)}, \end{aligned}$$

so by the weak formulation (V), $u'(1) = 0$. $\square$

Question 5. For $\Omega \subseteq \mathbb{R}^d$, $f \in L^2(\Omega)$ derive the variational formulation of the elliptic problem

$$\begin{cases} -\operatorname{div}_{\mathbf{x}}(c(\mathbf{x})\nabla u(\mathbf{x})) = f(\mathbf{x}), & \text{for } \mathbf{x} \in \Omega, \\ u = 0, & \text{on } \partial\Omega, \end{cases}$$

with $0 < c_{\min} \leq c(\mathbf{x}) \leq c_{\max}$ for all $\mathbf{x} \in \Omega$. Using Lax-Milgram's theorem, show that the weak formulation has a unique solution $u \in H_0^1(\Omega)$.

Solution 5. Integrate by parts against a test function $v \in H_0^1(\Omega)$,

$$\int_{\Omega} c(\mathbf{x})\nabla u(\mathbf{x}) \cdot \nabla v(\mathbf{x}) \, d\mathbf{x} = - \int_{\Omega} \operatorname{div}_{\mathbf{x}}[c(\mathbf{x})\nabla u(\mathbf{x})] v(\mathbf{x}) \, d\mathbf{x} = \int_{\Omega} f(\mathbf{x})v(\mathbf{x}) \, d\mathbf{x},$$

so leaving out the middle term we just obtained the weak formulation. The RHS defines a bounded linear functional on $H_0^1(\Omega)$ using Hölder's inequality. We now check that

$$(3) \quad a(u, v) = \int_{\Omega} c(\mathbf{x})\nabla u(\mathbf{x}) \cdot \nabla v(\mathbf{x}) \, d\mathbf{x}$$

satisfies the assumptions of the Lax-Milgram-theorem. Obviously a is bilinear and symmetric and by

$$\left| \int_{\Omega} c(\mathbf{x})\nabla u(\mathbf{x}) \cdot \nabla v(\mathbf{x}) \, d\mathbf{x} \right| \stackrel{\text{Hölder}}{\leq} c_{\max} \|\nabla u\|_{L^2} \|\nabla v\|_{L^2} \leq c_{\max} \|u\|_{H^1} \|v\|_{H^1}$$

it is also continuous. Moreover, using Poincaré's inequality,

$$a(v, v) \geq c_{\min} \int_{\Omega} |\nabla v|^2 \, d\mathbf{x} \gtrsim_{\Omega, c_{\min}} \|v\|_{L^2}^2$$

so a is coercive. Thus the assertion follows by applying the Lax-Milgram theorem. $\square$

Question 6. For $\Omega \subseteq \mathbb{R}^d$ open domain, $f \in L^2(\Omega)$ derive the variational formulation (V) of the elliptic problem

$$\begin{cases} -\operatorname{div}_{\mathbf{x}}(c(\mathbf{x})\nabla u(\mathbf{x})) = f(\mathbf{x}), & \text{for } \mathbf{x} \in \Omega, \\ u = 0, & \text{on } \partial\Omega, \end{cases}$$

with $0 < c_{\min} \leq c(\mathbf{x}) \leq c_{\max}$ for any $\mathbf{x} \in \Omega$. Then derive the Galerkin FE formulation and the resulting linear system $\mathbf{A}\mathbf{u} = \mathbf{f}$. What are the properties of the matrix $\mathbf{A}$?

Solution 6. Define a by (3) and $\ell(v) = \int_{\Omega} f v \, d\mathbf{x}$. We triangulate the domain Ω with a suitable triangulation $\mathcal{T}_h$. Then we define

$$V_h := \{v \in H_0^1 \mid v|_K \text{ is a polynomial of degree } r \, \forall K \in \mathcal{T}_h\}.$$

For simplicity choose $r = 1$. We replace our problem by the following approximation:

$$(4) \quad \text{find } u_h \in V_h \text{ such that } a(u_h, v) = \ell(v) \quad \text{for all } v \in V_h.$$

By linearity it suffices to consider a basis $\varphi_1, \dots, \varphi_n$ of V_h (by the way the finite-dimensional problem also has a unique solution because of Lax-Milgram). We write u_h and v with respect to that basis:

$$u = \sum_{i=1}^n u_i \varphi_i, \quad v = \sum_i v_i \varphi_i.$$

Plugging this into (4) and choosing $v = \varphi_i$ for all $i = 1, \dots, n$ gives a system of n equations i.e., we get

$$\mathbf{A}\mathbf{u} = \mathbf{f}, \quad \text{where } \mathbf{A} = (a(\varphi_i, \varphi_j))_{i,j=1}^n, \mathbf{f} = (\ell(\varphi_i))_{i=1}^n,$$

with $\mathbf{u} = (u_i)_{i=1}^n$.

We need to choose the basis functions φ_i , $i = 1, \dots, n$. Usually one takes a Lagrangian basis, that is a basis satisfying

$$\varphi_j(x_i) = \delta_{i,j} \quad \text{for all } x_i \text{ edge point of a triangle } K \in \mathcal{T}_h.$$

As a consequence, one often ends up with functions that are only supported in the triangles neighbouring the midpoint x_j . For example in 1D we can take the triangulation (x_i, x_{i+1}) for some x_i . Then the basis functions could be

$$\varphi_j(x) = \begin{cases} 1 - \frac{|x-x_j|}{h}, & x_{j-1} \leq x \leq x_{j+1} \\ 0, & \text{else,} \end{cases}$$

where $x_i - x_{i-1} = h$.

This ensures that we have to compute the integrals only over the support of φ_j . In higher dimensions it is *very* cumbersome to check if a basis function is zero on a given triangle for all triangles and all basis functions. To do this we decompose $a(u, v)$ and $\ell(v)$ in element wise contributions:

$$a_K(u, v) := \int_K c(\mathbf{x}) \nabla u(\mathbf{x}) \cdot \nabla v(\mathbf{x}) \, d\mathbf{x}$$

and analogously for ℓ so we have

$$a(u, v) = \sum_{K \in \mathcal{T}_h} a_K(u, v), \quad \ell(v) = \sum_{K \in \mathcal{T}_h} \ell_K(v).$$

This can be computed independently and in parallel: Fix $\hat{K} \in \mathcal{T}_h$. We will call this element in the following *reference triangle*. For each $K \in \mathcal{T}_h$ define

$$\mathbf{F}_{K_j} : \hat{K} \rightarrow K, \quad \boldsymbol{\xi} \mapsto \mathbf{P}_0 + \mathbf{J}_K \boldsymbol{\xi}.$$

After transforming to the reference element, we have basis functions $N_1, \dots, N_k \in \hat{K}[X]$ of the same degree as the basis functions φ_i . Then the $k \times k$ element stiffness matrix A_K for $K \in \mathcal{T}_h$ is defined for $i, j = 0, \dots, k$ (local numbering) by

$$\begin{aligned} (\mathbf{A}_K)_{i,j} &= a_K(N_i(\mathbf{F}_K^{-1}(\cdot)), N_j(\mathbf{F}_K^{-1}(\cdot))) \\ &= \int_K c(\mathbf{x}) \nabla_{\mathbf{x}} [N_i(\mathbf{F}_K^{-1}(\mathbf{x}))] \cdot \nabla_{\mathbf{x}} [N_j(\mathbf{F}_K^{-1}(\mathbf{x}))] \, d\mathbf{x} \\ &= \int_{\hat{K}} c(\mathbf{F}_K(\boldsymbol{\xi})) (D_{\boldsymbol{\xi}} N_i(\boldsymbol{\xi}) \cdot \mathbf{J}_K^{-1}) \cdot (D_{\boldsymbol{\xi}} N_j(\boldsymbol{\xi}) \mathbf{J}_K^{-1}) |\det \mathbf{J}_K| \, d\boldsymbol{\xi}. \end{aligned}$$

In order to implement this we need a function, mapping the local indices onto global ones. Note that the matrix $\mathbf{A}$ is symmetric and positive definite. The symmetry follows immediately by the symmetry of a . Moreover let $\boldsymbol{\xi} \in \mathbb{R}^N \setminus \{\mathbf{0}\}$. Then we have that

$$\begin{aligned} \boldsymbol{\xi}^\top \mathbf{A} \boldsymbol{\xi} &= \sum_{i,j=1}^N \xi_i a(\varphi_i, \varphi_j) \xi_j = a \left(\sum_{i=1}^N \xi_i \varphi_i, \sum_{j=1}^N \xi_j \varphi_j \right) \\ &\geq \alpha \left\| \sum_{i=1}^N \xi_i \varphi_i \right\|_{L^2} =: \|u_h\|_{L^2} \end{aligned}$$

Now, note that $u_h = \sum_{i=1}^N \xi_i \varphi_i$ is a continuous function. Since $\boldsymbol{\xi} \neq \mathbf{0}$ we have that there is an $i = 1, \dots, N$ such that $\xi_i \neq 0$. Then we have that $u_h(x_i) = \xi_i \varphi_i(x_i) = \xi_i \neq 0$. With the continuity this gives us that $\|u_h\|_{L^2} > 0$. □

Question 7. State and prove Céa's Lemma. Why is it important for the convergence theory of conforming FEM?

Solution 7.

Lemma 7.1 (Céa). Let $u \in V$ be the unique solution to

$$a(u, v) = \ell(v) \quad \text{for all } v \in V$$

and let $u_h \in V_h$ be the corresponding Galerkin-approximation. Then, the error satisfies

$$\|u - u_h\|_V \leq \frac{C}{\alpha} \min_{v \in V_h} \|u - v\|_V,$$

where α and C are the coercivity and continuity constants of a , both independent of V_h .

Proof. We have

$$\begin{aligned} a(u, v) &= \ell(v) \quad \text{for all } v \in V \text{ and} \\ a(u_h, v) &= \ell(v) \quad \text{for all } v_h \in V_h \subseteq V. \end{aligned}$$

Subtracting these two equations gives

$$(5) \quad a(u - u_h, v) = 0 \quad \text{for all } v \in V_h.$$

We call this last display *Galerkin-orthogonality*. This implies that we have for $v \in V_h$, using coercivity and continuity of a ,

$$\begin{aligned} \alpha \|u - u_h\|_V^2 &\leq a(u - u_h, \underbrace{u - u_h}_{=u-v+v-u_h}) = a(u - u_h, u - v) + \underbrace{a(u - u_h, v - u_h)}_{=0 \text{ by (5)}} \\ &\leq C \|u - u_h\|_V \|u - v\|_V. \end{aligned}$$

Since this result holds for any $v \in V_h$ we replace $\|u - v\|_V$ by its minimum and conclude.

This result is important because we now can choose a v that makes it easy to estimate $\|u - v\|_V$ like the P^1 interpolant. $\square$

Question 8. *Delineate the main steps in the FE convergence theory for 2D elliptic problems w.r.t. the H^1 -norm.*

Solution 8. The goal is to prove

$$\|u - u_h\|_{H^1} \lesssim_{\mathcal{T}} h \|u\|_{H^2}, \quad \text{for all } u \in H^2(\Omega).$$

The idea here is to use an interpolant of u to estimate the error.

Step 1 (Estimate on the reference element $\hat{K}$). Note that if $u \in H^2(\hat{K})$ then u can be represented by a function in $C^0(\bar{\hat{K}})$ by Sobolev's embedding theorem. For $\mathbf{z}_0, \dots, \mathbf{z}_2 \in \bar{\hat{K}}$ pairwise different and not co linear (not on one line) the interpolant

$$\begin{aligned} I_{\hat{K}} : H^2(\hat{K}) &\rightarrow \mathcal{P}^1(\hat{K}) \\ u &\mapsto [I_{\hat{K}} u : z_i \mapsto u(z_i)]. \end{aligned}$$

Proposition 8.1. *Let $u \in H^2(\hat{K})$ and $\mathbf{z}_0, \dots, \mathbf{z}_2, I_{\hat{K}}$ as above. Then*

$$\|u - I_{\hat{K}} u\|_{H^2(\hat{K})} \lesssim_{\hat{K}, \mathbf{z}_0, \dots, \mathbf{z}_2} |u|_{H^2(\hat{K})}.$$

Proof. Define the norm

$$\|v\| := \|v\|_{H^2} + \sum_{i=0}^2 |v(\mathbf{z}_i)|.$$

We claim that

$$(6) \quad \|v\| \gtrsim \|v\|_{H^2} \quad \text{for all } v \in H^2(\hat{K}).$$

By contradiction assume that there exists a sequence v_k with $k \|v_k\| < \|v_k\|_{H^2} = 1$. Thus the sequence v_k is bounded in $H^2(\hat{K})$ and thus has a in $H^1(\hat{K})$ convergent subsequence by Rellich' compactness. Assume without loss of generality that the whole sequence converges. Then

$$\begin{aligned} \|v_k - v_\ell\|_{H^2}^2 &\leq \|v_k - v_\ell\|_{H^1}^2 + \underbrace{2|v_k|_{H^2}^2 + |v_\ell|_{H^2}^2}_{\leq \|v_k\| \leq \frac{1}{k}} \xrightarrow{k, \ell \rightarrow +\infty} 0, \end{aligned}$$

i.e., v_k is Cauchy in H^2 and thus converges to some v^* . But

$$\|v^*\|_{H^2} = \lim_{k \rightarrow +\infty} \|v_k\|_{H^2} = 1 \quad \text{and} \quad \|v^*\| = \lim_{k \rightarrow +\infty} \|v_k\| \leq \lim_k \frac{1}{k} = 0,$$

Contradiction. This proves (6). The result follows from

$$\begin{aligned} \|u - I_{\hat{K}}u\|_{H^2(\hat{K})} &\stackrel{(6)}{\lesssim} \|u - I_{\hat{K}}u\| = |u - I_{\hat{K}}u|_{H^2} + \sum_{i=0}^2 |u - I_{\hat{K}}u(\mathbf{z}_i)| \\ &= |u|_{H^2} + \sum_{i=0}^2 0 = |u|_{H^2}. \end{aligned}$$

Step 2 (Extend to arbitrary triangle).

Lemma 8.1 (Scaling Lemma). *Let $K, \hat{K}$ be two triangles in $\mathbb{R}^2$ and let*

$$\begin{aligned} \mathbf{F}_K : \hat{K} &\rightarrow K \\ \boldsymbol{\xi} &\mapsto \mathbf{P} + \mathbf{J}_K \boldsymbol{\xi} = \mathbf{x} \end{aligned}$$

be such that $\mathbf{F}_K(\hat{K}) = K$ is an affine map. Then for $v \in H^m(K)$, $\hat{v}(\boldsymbol{\xi}) = v(\mathbf{F}_K(\boldsymbol{\xi})) = v(\mathbf{x})$ it holds that

$$|\hat{v}|_{H^m(\hat{K})} \lesssim_m \|\mathbf{J}_K\|_2^m |\det \mathbf{J}_K|^{-1/2} |v|_{H^m(K)}, \quad m = 0, 1, 2.$$

Proof. The result follows from taking the change of variables formula with

$$|\hat{v}|_{H^m(\hat{K})} = \int_K \sum_{|\alpha|=m} |\partial_{\boldsymbol{\xi}}^{\alpha} \hat{v}(\underbrace{\mathbf{F}_K^{-1}(\mathbf{x})}_{=\boldsymbol{\xi}})|^2 |\det D_{\mathbf{x}} \mathbf{F}_K^{-1}| d\mathbf{x}$$

Then one estimates the derivatives $|\partial_{\boldsymbol{\xi}}^{\alpha} \hat{v}(\boldsymbol{\xi})|$ using Cauchy-Schwartz.

In the following we will need the the definitions of h_K, ρ_K, h and the definitions of κ -shape regularity and quasi-uniformity.

Step 3 (Approximation error on $\mathcal{T}_h$).

Proposition 8.2. *Let $\Omega \subseteq \mathbb{R}^2$ be a polygonal domain, $\mathcal{T}_h$ a shape regular triangularisation with mesh size $h \leq h_0$ and I_h the piecewise interpolant.*

For $u \in H^2(\Omega)$ the approximation error satisfies

$$\|u - I_h u\|_{H^m(\Omega)} \lesssim h^{2-m} |u|_{H^2(\Omega)}, \quad m = 0, 1.$$

Proof. Let $K \in \mathcal{T}_h$, $m = 0, 1$. Then by the scaling-lemma, writing $\hat{u}(\boldsymbol{\xi}) = u(\mathbf{F}_K(\boldsymbol{\xi}))$

$$\begin{aligned} |u - I_h u|_{H^m(K)} &\lesssim_m |\mathbf{J}_K^{-1}|_2^m |\det \mathbf{J}_K|^{1/2} |\hat{u} - I_{\hat{K}} \hat{u}|_{H^m(\hat{K})}, \quad \text{by Lemma 8.1} \\ &\lesssim_{m, \hat{K}} |\mathbf{J}_K^{-1}|_2^m |\det \mathbf{J}_K|^{1/2} |\hat{u}|_{H^2(\hat{K})}, \quad \text{by Proposition 8.1} \\ &\lesssim_{m, \hat{K}} \|\mathbf{J}_K^{-1}\|_2^m |\det \mathbf{J}_K|^{1/2} \|\mathbf{J}_K\|_2^2 |\det \mathbf{J}_K|^{-1/2} |u|_{H^2(K)}, \text{ by Lemma 8.1} \\ &\leq \underbrace{\left(\frac{\hat{h}}{\hat{\rho}} \frac{h_K}{\rho_K} \right)^m}_{\leq \kappa} \frac{1}{\hat{\rho}^{2-m}} h_K^{2-m} |u|_{H^2(K)}. \end{aligned}$$

Summing over all triangles (careful $I_h \notin H^2(\Omega)$ but in $H^2(K)$ for each K !) we obtain the result.

Using all the previous results one obtains the convergence of the FEM in 2D. $\square$

Question 9. *State the 2D FEM L^2 -error estimate for elliptic problems and give the main steps in the proof (Aubin-Nitsche trick).*

Solution 9. First we have the following:

Proposition 9.1 (Elliptic regularity). *let $\Omega \subseteq \mathbb{R}^2$ be a bounded domain and $u \in H_0^1(\Omega)$ be the unique solution to (7) below. If $c \in C^1(\bar{\Omega})$ and if either*

(1) Ω is convex OR

(2) $\partial\Omega$ is C^2

then $u \in H^2(\Omega)$ and $\|u\|_{H^2(\Omega)} \lesssim_\Omega \|f\|_{L^2(\Omega)}$.

Consider the problem

$$(7) \quad \begin{cases} -\operatorname{div}_{\mathbf{x}}(c(\mathbf{x})\nabla u(\mathbf{x})) = f(\mathbf{x}) & \text{for } \mathbf{x} \in \Omega \\ u = 0 & \text{on } \partial\Omega \end{cases}$$

with $0 < \alpha \leq c(\mathbf{x}) \leq C$. We assume H^2 -regularity: The variational formulation

$$\text{find } u \in H_0^1(\Omega) \text{ such that } a(u, v) = \langle f, v \rangle_{L^2(\Omega)} \quad \text{for all } v \in H_0^1(\Omega)$$

with

$$a(u, v) = \int_{\Omega} c(\mathbf{x}) \nabla u(\mathbf{x}) \cdot \nabla v(\mathbf{x}) \, d\mathbf{x}.$$

Now, from the previous discussion we have for $u \in H^{r+1}(\Omega)$ that $\|u - u_h\|_{H^1} \lesssim h^r |u|_{H^{r+1}}$ but we want $\mathcal{O}(h^{r+1})$ (because of the higher regularity).

Proposition 9.2 (Aubin-Nitsche trick, duality). *Let $u \in H_0^1 \cap H^{r+1}(\Omega)$ be the unique solution to the above variational formulation under the full H^2 -regularity assumption (Proposition 9.1) above and $u_h \in V_h$ its Galerkin approximation. Then, the following L^2 -error estimate holds:*

$$\|u - u_h\|_{L^2(\Omega)} \lesssim h^{r+1} |u|_{H^{r+1}(\Omega)}.$$

Proof. Let $e = u - u_h \in H_0^1(\Omega) \subseteq L^2(\Omega)$ be the error and $\varphi \in H_0^1 \cap H^2(\Omega)$ be the unique solution to the dual problem

$$(8) \quad \begin{cases} -\operatorname{div}_{\mathbf{x}}(c(\mathbf{x})\nabla \varphi(\mathbf{x})) = e(\mathbf{x}) & \text{for } \mathbf{x} \in \Omega \\ \varphi = 0 & \text{on } \partial\Omega \end{cases}$$

By the regularity assumption, Proposition 9.1,

$$(9) \quad \|\varphi\|_{H^2(\Omega)} \lesssim \|e\|_{L^2(\Omega)}$$

Using that $e \in H_0^1$ we also have $a(\varphi, e) = (e, e)$. Next, denote by I^h the polynomial interpolant of degree r (careful φ is only in H^2). Then,

$$\begin{aligned} \|e\|_{L^2(\Omega)}^2 &= (e, e) = a(\varphi, e) = a(e, \varphi) \\ &= a(e, \varphi - I_h \varphi), \quad \text{by Galerkin-Orthogonality} \\ &\lesssim \|e\|_{H^1} \|\varphi - I_h \varphi\|_{H^1}, \quad \text{by continuity,} \\ &\leq \|e\|_{H^1} h \underbrace{\|\varphi\|_{H^2}}_{\leq \|\varphi\|_{H^2}}, \quad \text{by Proposition 8.2} \\ &\stackrel{(9)}{\lesssim} \|e\|_{H^1(\Omega)} h \|e\|_{L^2(\Omega)} \\ &\lesssim h^r |u|_{H^{r+1}} h \|e\|_{L^2}, \quad \text{by the convergence of FEM.} \end{aligned}$$

□

Question 10. *Consider the FE solution of the Poisson problem in 2D with homogeneous Dirichlet boundary conditions. What is a posteriori error estimation? Define the weak residual and state the upper error bound. What are the local error indicators? List the four main steps of the adaptive refinement algorithm.*

Solution 10. An a posteriori estimate is an estimate that only be done after the analysis and not before. The problem with our a priori estimates

$$\|u - u_h\|_{H^1} \lesssim h^r |u|_{H^{r+1}}$$

is that neither $|u|_{H^{r+1}}$ nor the constants in front of the RHS are known. How can we find estimates that include the numerical solution?

Let $\Omega \subseteq \mathbb{R}^2$ be a polygonal domain, $\mathcal{T}_h$ a shape-regular triangulation and

$$V_h = \{v \in C^0(\overline{\Omega}) \mid v|_K \in \Pi^1(K) \forall K \in \mathcal{T}_h, v|_{\partial\Omega} \equiv 0\}$$

and consider the Dirichlet problem

$$(10) \quad a(u, v) = \ell(v), \quad a(u, v) = \langle \nabla u, \nabla v \rangle_{L^2}, \quad \ell(v) = \langle v, f \rangle_{L^2},$$

for all $v \in V := H_0^1(\Omega)$.

If we insert u_h into the above equation we see that we have to minimize the residual

$$R(v) = \ell(v) - a(u_h, v) = a(u - u_h, v) = \begin{cases} 0 & \text{if } v \in V_h \\ a(u - u_h, v) & \text{else.} \end{cases}$$

Problem: In higher dimensions $d \geq 2$ we cannot use the standard nodal interpolant I_h , because H^1 -functions are not always continuous.

Definition. For $K \in \mathcal{T}_h$ we define

- $\mathcal{E}(K)$ as the set of edges of K in the interior of Ω .
- $\mathcal{N}(K)$ as the set of nodes/vertices of K .
- $\mathcal{E}(\mathcal{T}_h) := \bigcup_{K \in \mathcal{T}_h} \{e \in \mathcal{E}(K) \mid e \not\subseteq \partial\Omega\}$ as the interior edges of Ω ;
- $\mathcal{N}(\mathcal{T}_h) := \bigcup_{K \in \mathcal{T}_h} \mathcal{N}(K)$;
- $\omega_K := \bigcup_{K' : \mathcal{E}(K) \cap \mathcal{E}(K') \neq \emptyset} K'$;
- $\omega_e := \bigcup_{K \in \mathcal{T}_h : e \in \mathcal{E}(K)} K'$;
- $\tilde{\omega}_K := \bigcup_{\mathcal{N}(K) \cap \mathcal{N}(K') = \emptyset} K'$.
- $\tilde{\omega}_e := \bigcup_{\mathcal{N}(e) \cap \mathcal{N}(K') = \emptyset} K'$.

Define the *Clément interpolant* $I_h^C : H_0^1(\Omega) \rightarrow V_h$ in two steps:

- (1) For $x_j \in \mathcal{N}(\mathcal{T}_h) \cap \Omega$ let $\Pi_j : L^2(\omega_j) \rightarrow \mathcal{P}^0(\omega_j)$ be the local L^2 -projection onto the constant functions on the patch $\omega_j := \omega_{x_j}$ that is

$$\|v - \Pi_j v\|_{L^2(\omega_j)} = \min_{q \in \mathcal{P}^0(\omega_j)} \|v - q\|_{L^2}.$$

By L^2 orthogonality (just test $v - \Pi_j v$ against $q \equiv 1$ and use that $\Pi_j v \equiv \text{const}$) this is equivalent to

$$\langle v - \alpha, 1 \rangle_{L^2} = 0, \quad \alpha = \frac{1}{|\omega_j|} \int_{\omega_j} v(\mathbf{x}) \, d\mathbf{x}.$$

So for $v \in H_0^1(\Omega)$ $\Pi_j v$ is defined at every node x_j as

$$\Pi_j v = \begin{cases} \int_{\omega_j} v(\mathbf{x}) \, d\mathbf{x}, & x_j \ni \partial\Omega \\ 0 & \text{else.} \end{cases}$$

- (2) Now interpolate all values $\Pi_j v$ at the nodes $\mathcal{N}(\mathcal{T}_h)$

$$I_h^C[v](x) := \sum_{x_j \in \mathcal{N}(\mathcal{T}_h)} (\Pi_j v) \varphi_j(x) \in V_h$$

where φ_j are the Lagrangian basis functions of V_h .

This interpolant satisfies the following:

Lemma 10.1. *Let $K \in \mathcal{T}_h$, $e \in \mathcal{E}(\mathcal{T}_h)$, $v \in H_0^1(\Omega)$ and $\{\mathcal{T}_h\}_h$ be κ -shape regular. Then,*

- (i) $\|v - I_h^C v\|_{L^2(K)} \leq c_1 h_K \|\nabla v\|_{L^2(\omega_K)},$
- (ii) $\|v - I_h^C v\|_{L^2(e)} \leq c_2 \sqrt{h_e} \|\nabla v\|_{L^2(\omega_e)},$

where h_e is the length of edge e and the constants c_1, c_2 only depend on the shape-regularity (on the largest interior angle) of $\{\mathcal{T}_h\}_h$.

We obtain for any $v \in V$ by integration by parts on each triangle K ,

$$\begin{aligned} R(v) &= \sum_{K \in \mathcal{T}_h} \left(\langle f, v \rangle_{L^2(K)} - \langle \nabla u_h, \nabla v \rangle_{L^2(K)} \right) \\ &= \sum_K \left(\langle f, v \rangle_{L^2(K)} + \underbrace{\langle \Delta u_h, v \rangle_K}_{=0 \text{ as } u_h|_K \in \mathcal{P}^1(K)} - \int_{\partial K} (\nabla u_h \cdot \mathbf{n}_K) v \, dS \right) \\ &= \sum_K \langle f, v \rangle_{L^2(K)} - \sum_{e \in \mathcal{E}(\mathcal{T}_h)} \int_e \left[\frac{\partial u_h}{\partial \mathbf{n}} \right]_e v \, dS. \end{aligned}$$

Here,

$$\left[\frac{\partial u_h}{\partial \mathbf{n}} \right]_e := \nabla u_h|_{K_1} \cdot \mathbf{n}_{K_1} + \nabla u_h|_{K_2} \cdot \mathbf{n}_{K_2} \stackrel{\mathbf{n}_{K_1} = -\mathbf{n}_{K_2}}{=} \left(\nabla u_h|_{K_1} - \nabla u_h|_{K_2} \right) \cdot \mathbf{n}_{K_1},$$

denotes the *jump in the normal derivative* across edge e .

We now prove the following:

Proposition 10.1. *Let $\{\mathcal{T}_h\}_h$ be a shape-regular triangulation of $\Omega \subseteq \mathbb{R}^2$ and u_h be the FE-solution to (10). Then*

$$\|\nabla(u - u_h)\|_{L^2(\Omega)} \leq C \left(\sum_{K \in \mathcal{T}_h} \eta_K^2 \right)^{1/2},$$

where

$$\eta_K^2 = h_K^2 \|f\|_{L^2(K)}^2 + \frac{1}{2} \sum_{e \in \mathcal{E}(K)} h_e \left\| \left[\frac{\partial u_h}{\partial \mathbf{n}} \right]_e \right\|_{L^2(e)}^2$$

and the constant C only depends on the shape regularity of the mesh.

Proof. Take $v \in V = H_0^1(\Omega)$. Since $I_h^C v \in V_h$ we have $R(I_h^C v) = 0$, so

$$R(v) = R(v - I_h^C v) = \sum_{K \in \mathcal{T}} \langle f, v - I_h^C v \rangle_{L^2(K)} - \sum_{e \in \mathcal{E}(\mathcal{T})} \int_e \left[\frac{\partial u_h}{\partial \mathbf{n}} \right]_e (v - I_h^C v) \, dS.$$

Now, using Cauchy-Schwartz and the above Lemma and then the discrete Cauchy-Schwartz we obtain

$$\begin{aligned} |R(v)| &\leq \sum_{K \in \mathcal{T}} \|f\|_{L^2(K)} \|v - I_h^C v\|_{L^2(K)} + \sum_{e \in \mathcal{E}(T)} \left\| \left[\frac{\partial u_h}{\partial \mathbf{n}} \right]_e \right\|_{L^2(e)} \|v - I_h^C v\|_{L^2(e)} \\ &\leq \sum_{K \in \mathcal{T}} \|f\|_{L^2(K)} c_1 h_K \|\nabla v\|_{L^2(\omega_K)} \\ &\quad + \sum_{e \in \mathcal{E}(T)} \left\| \left[\frac{\partial u_h}{\partial \mathbf{n}} \right]_e \right\|_{L^2(\omega_e)} c_2 \sqrt{h_e} \|\nabla v\|_{L^2(\omega_e)} \\ &\leq c_1 \left(\sum_{K \in \mathcal{T}} \|f\|_{L^2(K)}^2 h_K^2 \right)^{1/2} \left(\sum_{K \in \mathcal{T}} \|\nabla v\|_{L^2(\omega_K)}^2 \right)^{1/2} \\ &\quad + c_2 \left(\sum_{e \in \mathcal{E}(T)} h_e \left\| \left[\frac{\partial u_h}{\partial \mathbf{n}} \right]_e \right\|_{L^2(e)}^2 \right)^{1/2} \left(\sum_{e \in \mathcal{E}(T)} \|\nabla v\|_{L^2(\omega_e)}^2 \right)^{1/2} \\ &\leq C \left(\sum_{K \in \mathcal{T}} \eta_K^2 \right) \|\nabla v\|_{L^2(\Omega)} \end{aligned}$$

Substitute $v = u - u_h$ and conclude.

This gives us the following algorithm:

- (1) Determine the FE-Solution u_h for $\mathcal{T}^{(\ell)}$

- (2) For any $K \in \mathcal{T}^{(\ell)}$ determine η_K as above.
 If $(\sum_{K \in \mathcal{T}^{(\ell)}} \eta_K^2)^{1/2} < \text{tol}$ then stop; otherwise
- (3) Mark all $K \in \mathcal{T}^{(\ell)}$ that are "large".
- (4) Refine the marked triangles $\rightarrow$ new mesh $\mathcal{T}^{(\ell+1)}$ and go to 1.

One often uses either

- *Maximum strategy*: Mark all $K \in \mathcal{T}^{(\ell)}$ with $\eta_K \geq \vartheta \eta_{\max}$
- *Dörfler marking* Mark all $K \in \tilde{\mathcal{T}} \subseteq \mathcal{T}^{(\ell)}$ where the subset $\tilde{\mathcal{T}}$ satisfies

$$\sum_{K \in \tilde{\mathcal{T}}} \eta_K^2 \geq \vartheta \sum_{K \in \mathcal{T}^{(\ell)}} \eta_K^2.$$

□

Question 11. What are DG-FEM? For the 2D Poisson equation with homogeneous Dirichlet boundary conditions, derive the variational formulation. Which two terms need to be added in the SIP-DG formulation for the discrete problem to be well-posed?

Solution 11. Discontinuous-Galerkin-Finite Element Methods are finite elements where we drop the requirement of v_h being continuous. The advantage is that we then can also approximate discontinuous functions and so on. Problem: The previous theory does not hold.

We again consider the problem (2) with $f, g \equiv 0$. For simplicity assume $u \in H_0^1 \cap H^2(\Omega)$. Then

- (1) $u \in C^0(\bar{\Omega})$
- (2) $\nabla u \in H^1(\Omega)$ and $\nabla u|_{\partial K} \in L^2(\partial K, \mathcal{H}^{d-1})$ in the trace sense for all $K \in \mathcal{T}_h$.

Definition. Let $K_+, K_- \in \mathcal{T}_h$ with $e \in \partial K_+ \cap \partial K_- \neq \emptyset$. For $v : K_+ \cup K_- \rightarrow \mathbb{R}$ we denote $v_{\pm}$ the trace of v on e from $K_{\pm}$, i.e.,

$$v_{\pm} = \gamma(v|_{K_{\pm}})|_e$$

Then we define the *normal jump* by

$$[[v]]_e := v_1 \mathbf{n}_+ + v_- \mathbf{n}_-$$

and the *average* of v on e as

$$\{v\}_e := \frac{v_+ + v_-}{2}.$$

One can prove that $v \in H^1(\Omega)$ if $[[v]]_e \equiv 0$ a.e. for all edges e .

We now derive the weak formulation. Let $u \in H^2 \cap H_0^1(\Omega)$ be the solution to the problem. Multiply by a testfunction $v \in V_h$ and integrate over $K \in \mathcal{T}_h$,

$$\int_K (-\Delta u) v \, d\mathbf{x} = \int_K \nabla u \cdot \nabla v \, d\mathbf{x} - \int_{\partial K} \nabla u \cdot \mathbf{n}_K v \, dS = \int_K f v \, d\mathbf{x}.$$

Then sum over all $K \in \mathcal{T}_h$ to get

$$\sum_{K \in \mathcal{T}_h} \left(\int_K \nabla u \cdot \nabla v \, d\mathbf{x} - \int_{\partial K} \nabla u \cdot \mathbf{n}_K v \, dS \right) = \int_{\Omega} f v \, d\mathbf{x}$$

or equivalently, using $v|_{\partial\Omega} \equiv 0$,

$$\sum_{K \in \mathcal{T}_h} \int_K \nabla u \cdot \nabla v \, d\mathbf{x} - \sum_{e \in \mathcal{E}(\mathcal{T}_h)} \int_e [[(\nabla u)v]]_e \, dS = \int_{\Omega} f v \, d\mathbf{x}.$$

Unfortunately the bilinear form on the LHS is neither symmetric nor coercive on $V_h \times V_h$. We thus add

$$- \sum_{e \in \mathcal{E}(\mathcal{T}_h)} \int_e \{ \nabla v \}_e [[u]]_e \, dS + \sum_{e \in \mathcal{E}(\mathcal{T}_h)} \int_e \frac{\sigma_e}{|e|} [[u]]_e [[v]]_e \, dS$$

and obtain an updated bilinear form a_{DG} . First term guarantees symmetry and the second term penalizes the jumps with an interior penalty $\sigma_e > 0$.

Definition. For any $v \in V_h$ we define the mesh-dependent DG-energy norm by

$$\|v\|_h^2 := \sum_{K \in \mathcal{T}_h} \|\nabla v\|_{L^2(K)}^2 + \sum_{e \in \mathcal{E}(\mathcal{T}_h)} \frac{\sigma_e}{|e|} \|\llbracket v \rrbracket_e\|_{L^2(e)}^2$$

Proposition 11.1. *There exists a $\sigma_{\min} > 0$ only depending on the shape regularity κ of $\mathcal{T}_h$, such that if $\sigma_e \geq \sigma_{\min}$ for all $e \in \mathcal{E}(\mathcal{T}_h)$, then*

$$a_{\text{DG}}(v, v) \geq \alpha \|v\|_h^2 \quad \text{for all } v \in V_h$$

for some $\alpha > 0$ independent of h . So a_{DG} is coercive on $V_h \times V_h$ and thus the discrete DG formulation has a unique solution for $\sigma_{\min} > 0$ sufficiently large. $\square$

Question 12. *Derive the variational formulation of the heat equation in 2D with homogeneous Dirichlet boundary conditions. Then derive the semi-discrete Galerkin and the fully discrete formulations using the θ -method for time discretization.*

Solution 12. Consider the heat equation

$$\begin{cases} \partial_t u - \operatorname{div}(c \nabla u) = f & \text{on } (0, T) \times \Omega, \\ u = 0 & \text{on } (0, T) \times \partial\Omega \text{ and} \\ u(0, \cdot) = u_0 & \text{in } \Omega, \end{cases}$$

where $0 < c_{\min} < c(\mathbf{x}) < c_{\max}$ for all $\mathbf{x} \in \Omega$.

If u is sufficiently differentiable we can multiply with a testfunction $v \in C_0^\infty(\Omega)$ and obtain

$$\frac{d}{dt} \int_{\Omega} u(t, \mathbf{x}) v(\mathbf{x}) \, d\mathbf{x} + \underbrace{\int_{\Omega} c(\mathbf{x}) \nabla_{\mathbf{x}} u(t, \mathbf{x}) \cdot \nabla v(\mathbf{x}) \, d\mathbf{x}}_{=: a(u(t), v)} = \int_{\Omega} f(t, \mathbf{x}) v(\mathbf{x}) \, d\mathbf{x}.$$

The variational formulation then is: find $u \in C((0, T); L^2(\Omega)) \cap L^2((0, T); H_0^1(\Omega))$ such that

$$\begin{cases} \frac{d}{dt} \langle u(t), v \rangle_{L^2} + a(u(t), v) = \langle f(t), v \rangle_{L^2} & \text{for all } v \in H_0^1(\Omega), t \in (0, T) \\ u(0) = u_0 \end{cases}$$

Take again $V_h \subseteq H_0^1(\Omega)$ with $\dim V_h = N < +\infty$ and $\varphi_1, \dots, \varphi_N$ a Lagrangian basis of V_h . We approximate by the problem: find $u_h : [0, T] \rightarrow V_h$ such that

$$(11) \quad \begin{cases} \frac{d}{dt} \langle u_h(t), v \rangle_{L^2} + a(u_h(t), v) = \langle f(t), v \rangle_{L^2} & \text{for all } v \in H_0^1(\Omega), t \in (0, T) \\ u_h(0) = u_{0,h}. \end{cases}$$

Writing $u_h(t) = \sum_{j=1}^N u_j(t) \varphi_j$ we have that (11) is equivalent to

$$\begin{cases} \mathbf{M} \mathbf{U}'(t) + \mathbf{A} \mathbf{U}(t) = \mathbf{F}(t) & \text{for } t \in (0, T) \\ \mathbf{M} \mathbf{U}(0) = \mathbf{U}^0 \end{cases}$$

where

$$\begin{aligned} \mathbf{A} &= [a(\varphi_i, \varphi_j)]_{i,j=1}^N, \mathbf{M} = [\langle \varphi_i, \varphi_j \rangle]_{i,j=1}^N, \mathbf{U}(t) = \begin{bmatrix} u_i(t) \end{bmatrix}_{i=1}^N, \\ \mathbf{F}(t) &= [\langle f(t), \varphi_i \rangle]_{i=1}^N, \mathbf{U}^0 = [\langle u_{0,h}, \varphi_i \rangle]_{i=1}^N. \end{aligned}$$

Considering this we can take $t_m = m \cdot \Delta t$ take the following recursion

$$\mathbf{M} \frac{\mathbf{U}^{m+1} - \mathbf{U}^m}{\Delta t} = - \left(\vartheta \mathbf{A} \mathbf{U}^{m+1} + (1 - \vartheta) \mathbf{A} \mathbf{U}^m \right) + \vartheta \mathbf{F}^{m+1} + (1 - \vartheta) \mathbf{F}^m$$

for $\vartheta \in (0, 1)$. If $\vartheta = 1$ this corresponds to implicit Euler-method, $\vartheta = 0$ to the explicit one and $\vartheta = \frac{1}{2}$ to Crank-Nicolson (which converges with $\mathcal{O}(\Delta t^2)$ whereas Euler only converges with $\mathcal{O}(\Delta t)$). $\square$

APPENDIX A. VARIOUS DENSITY RESULTS, SOBOLEV INEQUALITY, ANALYSIS OF PDES, ETC.

Proposition 12.1. *Let $\Omega \subseteq \mathbb{R}$ be an open and bounded interval. If $u \in H^1(\Omega)$ then there exists $u \in C^0(\overline{\Omega})$ such that $u \equiv \tilde{u}$ a.e..*

Proposition 12.2. *Let $\Omega \subseteq \mathbb{R}^d$ be an open domain. Then $C^\infty \cap H^k(\Omega)$ is dense in $H^k(\Omega)$ for all $k \in \mathbb{N}_0$.*

Theorem 12.1 (Sobolev Embedding). *Let $\Omega \subseteq \mathbb{R}^d$ be a bounded Lipschitz domain. Then*

$$H^m(\Omega) \hookrightarrow C^k(\overline{\Omega}) \quad \text{for } m > k + \frac{d}{2}.$$

This theorem says that the identity map $H^m(\Omega) \rightarrow C^k(\overline{\Omega})$ is continuous. Since the identity is a linear operator we obtain

$$\|u\|_{C^k(\overline{\Omega})} \lesssim \|u\|_{H^m(\Omega)}, \quad \text{for all } u \in H^m(\Omega).$$

Proposition 12.3 (Poincaré-Friederich's). *Let $\Omega \subseteq \mathbb{R}^d$ be a bounded domain. Then*

$$\|v\|_{L^2} \lesssim \|\nabla v\|_{L^2} \quad \text{for all } v \in H_0^1(\Omega).$$

Proof. Let $u \in C_c^\infty(\Omega)$ which we regard as a function $u : \mathbb{R}^d \rightarrow \mathbb{R}$ by extending it trivially outside Ω . Moreover, after a suitable translation and rotation in $\mathbb{R}^d$ we can without loss of generality assume that

$$\Omega \subseteq (0, \underbrace{\text{diam}(\Omega)}_{=:L}) \times \mathbb{R}^{d-1}.$$

For $\mathbf{x} = (x_1, \mathbf{x}') \in \Omega$ we have

$$\begin{aligned} |u(x_1, x')|^2 &= \left| \int_0^{x_1} \frac{\partial u}{\partial x_1}(s, \mathbf{x}') \, ds \right|^2 \\ &\leq \left(\int_0^L \left| \frac{\partial u}{\partial x_1}(s, \mathbf{x}') \right| \, ds \right)^2 \\ &\leq \left(\int_0^L 1 \, ds \right) \cdot \int_0^L |\nabla u(s, \mathbf{x}')|^2 \, ds \\ &\leq L \int_0^L |\nabla u(s, \mathbf{x}')|^2 \, ds, \end{aligned}$$

where we used that $|\frac{\partial u}{\partial x_1}| \leq |\nabla u|$ a.e. and the Cauchy-Schwartz inequality. Thus

$$\begin{aligned} \int_{\Omega} |u|^2 \, d\mathbf{x} &\leq \int_{\mathbb{R}^{d-1}} \int_0^L |u(x_1, \mathbf{x}')|^2 \, dx_1 \, d\mathbf{x}' \\ &\leq \int_{\mathbb{R}^{d-1}} \int_0^L \left(L \int_0^L |\nabla u(s, \mathbf{x}')|^2 \, ds \right) \, dx_1 \, d\mathbf{x}' \\ &\leq \int_{\mathbb{R}^{d-1}} L^2 \int_0^L |\nabla u(s, \mathbf{x}')|^2 \, ds \, d\mathbf{x}' \\ &\leq L^2 \int_{\Omega} |\nabla u|^2 \, d\mathbf{x}. \end{aligned}$$

We conclude by density.

We can define the trace of a function which replaces boundary conditions.

Definition. Let $u \in C^1(\overline{\Omega})$ we call

$$\begin{aligned} \gamma : C^1(\overline{\Omega}) &\rightarrow C^0(\Gamma) \\ u &\mapsto \gamma u =: u|_{\Gamma} \end{aligned}$$

This trace can be extended to function in $H^1(\Omega)$.

Theorem 12.2 (Trace Theorem). *Let $\Omega \subseteq \mathbb{R}^d$ be a Lipschitz-domain with $\partial\Omega = \Gamma$ piecewise C^1 . Then $C^1(\bar{\Omega})$ lies densely in $H^1(\Omega)$ and the trace map γ can be extended continuously to a linear map*

$$\gamma : H^1(\Omega) \rightarrow L^2(\Gamma, \mathcal{H}^{d-1})$$

$$u \mapsto \gamma u.$$

APPENDIX B. IMPORTANT DEFINITIONS

Definition (Lipschitz domain). Let $\Omega \subseteq \mathbb{R}^d$ be a bounded domain with $d \geq 2$. Denote $\Gamma := \partial\Omega$. Ω is called *Lipschitz-domain* if for all $x_0 \in \Gamma$ there exists $\varepsilon > 0$ and $w \in \mathbb{R}^d$ with $|w| = 1$ and a Lipschitz-continuous map g such that

- (1) every $x \in \mathbb{R}^d$ can be uniquely written as $x = y + \xi w$, where $\xi \in \mathbb{R}$ and

$$y \in H := \left\{ y \in \mathbb{R}^d \mid y - x_0 \perp w \right\}.$$

- (2) For $H^\varepsilon := H \cap B_\varepsilon(x_0)$ there exists $g : H^\varepsilon \rightarrow \mathbb{R}$ Lipschitz-continuous and $\eta > 0$ such that for the cylinder $U_{\varepsilon, \eta} := H^\varepsilon + (-\eta, \eta)w$ it holds that

$$\Omega \cap U_{\varepsilon, \eta} = \{y + \xi w \mid y \in H^\varepsilon, -\eta < \xi < g(y)\}.$$

Definition. A triangulation $\mathcal{T} = \{K_j\}_{j \in J}$ is a partition of Ω into open disjoint triangles/quadrilateral with $\bar{\Omega} = \bigcup_j \bar{K}_j$. The triangulation $\mathcal{T}$ is called *regular* if for all $K_1, K_2 \in \mathcal{T}$

$$\bar{K}_1 \cap \bar{K}_2 = \begin{cases} \emptyset \\ \text{vertex or} \\ \text{entire edge.} \end{cases}$$

Definition. Let $\mathcal{T}$ be a regular triangulation of Ω . For $K \in \mathcal{T}$ we call

- $h_K :=$ radius of the circumscribed circle of K ;
- $\rho_K :=$ radius of the inscribed circle of K and
- $h = \max_{K \in \mathcal{T}} h_K$ the mesh size of h .

Definition. Let $\{\mathcal{T}_h\}_{h \in (0, h_0)}$ be a family of regular triangulations.

- $\{\mathcal{T}_h\}_h$ is called κ -shape regular if there exists $\kappa > 0$ such that

$$\max_{K \in \mathcal{T}_h} \frac{h_K}{\rho_K} \leq \kappa \quad \text{for all } \mathcal{T}_h \in \{\mathcal{T}_h\}_h.$$

- $\{\mathcal{T}_h\}_h$ is quasi-uniform if there exists $c > 0$ such that

$$\frac{\max_{K \in \mathcal{T}_h} h_K}{\min_{K \in \mathcal{T}_h} h_K} \leq c \quad \text{for all } \mathcal{T}_h \in \{\mathcal{T}_h\}_h.$$